

THE TRANSFORMATIVE POWER OF GEOAI: A CASE OF STUDY FOR SMALL FARMERS' EUDR COMPLIANCE

DIPLOMA THESIS
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RESEARCH PROBLEM

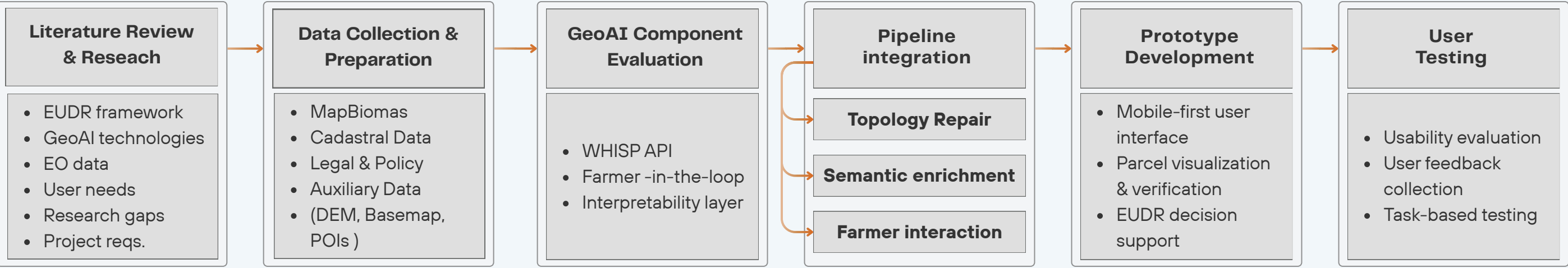
How can GeoAI-powered digital tools bridge the gap between EUDR compliance requirements and smallholder farmers' demands?

The EUDR requires plot-level geolocation and deforestation evidence. Smallholders often face inaccurate boundaries, fragmented data and limited technical capacity. This research connects automated spatial validation, Earth Observation and farmer knowledge through an explainable, user-in-the-loop GeoAI workflow.

OBJECTIVES

- 01 Design a replicable and scalable **farmer-in-the-loop GeoAI pipeline** for parcel harmonization and topology repair to support EUDR compliance for small-farmers.
- 02 Translate pipeline into an **interactive dashboard** integrating GeoAI to support parcel validation and deliver interpretable EUDR compliance insights.
- 03 Evaluate and document the system through technical testing, active-learning records, and structured user studies, assessing its usability and transparency.

METHODOLOGY



RESULTS - PIPELINE

